

Customer Segmentation Project for Smarter Marketing

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EXECUTIVE SUMMARY

This project analyzes transactional data from an online retail company, spanning December 2010 to December 2011, which contains 541,909 transactions from 4,339 unique customers across 38 countries. Using RFM (Recency, Frequency, Monetary) analysis together with K-means clustering, four distinct customer segments were identified in order to enable targeted marketing strategies and enhance customer lifetime value.

Analysing the data revealed significant variation in customer behavior. Champions (49.9% of customers) generate disproportionate revenue with an average expenditure of \$4,238 and 135 transactions, purchasing as recently as 21 days ago. At-risk customers (30.7%) show concern about disengagement patterns, averaging 98 days since last purchase with declining transaction frequency. Lost customers (19.1%) have been inactive for over 9 months and represent immediate churn. A small B2B segment (0.3%, 13 customers) was identified with purchasing patterns 63 times higher than retail customers, averaging \$201,789 in total expenditure.

Implementation of segment-specific strategies could yield substantial business impact. Re-engagement campaigns targeting 'At-Risk' customers could recover an estimated 400-450 customers worth \$310,000-400,000 annually. Win-back campaigns for 'Lost' customers could recapture 80-125 customers worth \$50,000-80,000 annually. Proper B2B account management could increase retention and expand this high-value segment. Total potential revenue recovery is estimated at \$360,000-480,000 within 12 months."

INTRODUCTION

Business Context

In today's highly competitive retail landscape, acquiring new customers costs 5-25 times more than retaining existing ones (Gallo, 2014). Yet many companies allocate disproportionate resources to acquisition while neglecting their current customer base. The key to sustainable growth lies not just in attracting customers but in understanding their behaviors, predicting their needs, and tailoring marketing strategies accordingly.

Customer segmentation allows businesses to move beyond one-size-fits-all marketing toward personalized experiences that resonate with distinct customer groups. By identifying which customers are highly engaged, which are at risk of churning, and which have already left, organizations can allocate resources efficiently and maximize return on marketing investment.

Problem Statement

This online retail company faces several critical challenges:

1. **Lack of customer visibility:** With thousands of transactions monthly, understanding individual customer behavior at scale is impossible without systematic analysis.
2. **Inefficient marketing spending:** Generic promotional campaigns reach all customers equally, wasting resources on those unlikely to respond while under-investing in high-value segments.
3. **Silent churn:** Customers stop purchasing without warning, and the company only recognizes churn retrospectively when it's too late for intervention.
4. **Undifferentiated service:** All customers receive identical treatment despite vast differences in value, loyalty, and needs.
5. **Revenue leakage:** Without understanding customer lifecycle stages, the company misses opportunities for retention, upselling, and win-back.

The fundamental question this analysis addresses is: How can we systematically segment our customer base to enable targeted, cost-effective marketing strategies that maximize customer lifetime value and minimize churn?

Objectives

This project aims to:

1. **Segment customers** into distinct groups based on purchasing behavior using data-driven methods
2. **Characterize each segment** with clear behavioral profiles (recency, frequency, monetary patterns)
3. **Quantify segment sizes** and their contribution to overall revenue
4. **Identify at-risk customers** showing early warning signs of churn
5. **Develop actionable strategies** tailored to each segment's characteristics and needs
6. **Estimate business impact** of implementing segment-specific marketing campaigns
7. **Establish a framework** for ongoing customer intelligence and monitoring

Approach Overview

This analysis employs RFM (Recency, Frequency, Monetary) analysis combined with K-means clustering - a proven methodology for customer segmentation in retail contexts. RFM analysis reduces complex transactional data to three interpretable behavioral dimensions:

- **Recency:** How recently did the customer purchase? (Indicates engagement level)
- **Frequency:** How often does the customer purchase? (Indicates loyalty and habit formation)
- **Monetary:** How much does the customer spend? (Indicates economic value)

K-means clustering then identifies natural groupings within the customer base based on these three dimensions, revealing segments with distinct behavioral patterns requiring different marketing approaches.

Dataset Overview

The analysis uses transactional data from a UK-based online retailer specializing in unique all-occasion gifts. The dataset spans 13 months (December 2010 - December 2011) and includes:

- **541,909 transactions** across 4,070 unique product SKUs
- **4,372 unique customers** (after removing missing IDs)

- **38 countries** represented, with UK comprising the majority
- **Transaction details:** Invoice numbers, dates, quantities, unit prices, product descriptions
- **Time period:** Covers the full annual cycle, including peak holiday season

The company operates primarily online (e-commerce) with some wholesale business evident in the data. Products range from decorative items and gifts to household goods, with unit prices from \$0.001 to \$38,970.

DATA EXPLORATION & PREPARATION

Dataset Structure: The raw dataset contained 541,909 records with 8 variables:

- InvoiceNo: Unique transaction identifier
- StockCode: Product SKU
- Description: Product name
- Quantity: Number of units purchased
- InvoiceDate: Transaction timestamp
- UnitPrice: Price per unit in GBP
- CustomerID: Unique customer identifier
- Country: Customer location

Data Quality Issues Identified:

1. **Missing CustomerIDs:** 135,080 records (24.9%) lacked customer identifiers, preventing attribution to specific customers. These represent guest checkouts or possible data collection errors.
2. **Negative Quantities:** 8,905 records showed negative quantities, indicating product returns or cancellations rather than purchases.
3. **Zero/Negative Prices:** Some records show \$0.00 or negative prices, suggesting data entry errors, discounts, or adjustments.
4. **Data Type Issues:** InvoiceDate stored as text rather than datetime, requiring conversion.

Data Cleaning Decisions

Removed Records:

- Missing CustomerIDs (135,080)
- Negative quantities (8,905)
- Final clean dataset: 397,924 transactions from 4,339 customers

Retained Issues:

- Kept all price points including extreme values (\$0.002 - \$62,352) to observe natural customer segmentation
- Retained all countries despite UK dominance (82%) to capture international customer patterns
- Kept extremely high-value orders to identify wholesale/B2B customers

Exploratory Analysis Findings

Temporal Patterns:

- Dataset spans December 1, 2010 to December 9, 2011 (374 days)
- Peak transaction periods: November-December (holiday season)
- Lowest activity: January-February (post-holiday slump)
- Average transactions per day: 1,063

Customer Distribution:

- Total unique customers: 4,339
- Average transactions per customer: 91.7
- Median transactions per customer: 41
- Range: 1 to 7,847 transactions (indicating wide behavioral variance)

Revenue Patterns:

- Total revenue (cleaned data): ~\$14.2 million
- Average customer value: \$3,285
- Median customer value: \$1,078
- Top 10% of customers contribute ~60% of revenue (Pareto principle confirmed)

Geographic Distribution:

- United Kingdom: 82% of customers

- Germany: 3%
- France: 2.5%
- EIRE (Ireland): 2.3%
- Other 34 countries: 10.2%

Product Insights:

- 4,070 unique products
- Most purchased categories: Decorative items, gifts, party supplies
- Average order size: 9.6 items
- Price distribution: Highly right-skewed with long tail of high-value items

Key Observations:

1. Customer behavior shows extreme variance - some purchase once, others thousands of times
2. Clear presence of both B2C (retail) and B2B (wholesale) customers
3. Strong seasonality suggests need for time-based segmentation in future analysis
4. International customers represent growth opportunity but require localized strategies

Feature Engineering: RFM Metrics

Recency Calculation:

- Reference date: December 10, 2011 (one day after last transaction)
- Formula: (Reference date - Customer's most recent purchase date)
- Range: 1 to 374 days
- Distribution: Right-skewed with mode around 20-30 days

Frequency Calculation:

- Metric: Count of unique invoice numbers per customer
- Range: 1 to 7,847 transactions
- Mean: 91.7, Median: 41
- Distribution: Heavily right-skewed with extreme outliers

Monetary Calculation:

- Formula: Sum of (Quantity × UnitPrice) for all customer transactions
- Range: \$6.00 to \$448,330
- Mean: \$3,285, Median: \$1,078

- Distribution: Highly right-skewed

RFM Summary Statistics:

Table 1: RMF Summary Statistics

Metric	Mean	Median	Std Dev	Min	Max
Recency (days)	92.5	51	100.0	1	374
Frequency (orders)	91.7	41	228.8	1	7,847
Monetary (\$)	3,285	1,078	14,381	6.00	448,330

Note: All monetary values converted from GBP to USD using exchange rate of £1 = \$1.60 (December 2010-2011 average)

Correlation Analysis:

- Recency vs Frequency: -0.28 (weak negative - recent customers tend to purchase more frequently)
- Recency vs Monetary: -0.18 (weak negative - recent customers tend to spend more)
- Frequency vs Monetary: 0.76 (strong positive - frequent buyers spend more total)

The strong frequency-monetary correlation suggests these dimensions aren't entirely independent, but both provide valuable segmentation information.

Outlier Analysis

Using the Interquartile Range (IQR) method with a $3 \times \text{IQR}$ threshold for extreme outliers:

Monetary Outliers:

- Q1: \$491, Q3: \$2,659, IQR: \$2,168
- Upper bound: $\$2,659 + 3(\$2,168) = \$9,163$
- Identified: 222 customers (5.1%) exceeding this threshold
- These customers spend \$9,163 to \$448,330 total

Frequency Outliers:

- Customers with >500 transactions: 47 customers
- Maximum frequency: 7,847 transactions (likely wholesale/reseller)

All the outliers have been kept during clustering to observe natural segmentation. These extreme customers either form their own cluster (B2B segment) or influence cluster boundaries in meaningful ways.

Frequency Outliers:

- Customers with >500 transactions: 47 customers
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All the outliers have been kept during clustering to observe natural segmentation. These extreme customers either form their own cluster (B2B segment) or influence cluster boundaries in meaningful ways.

METHODOLOGY

Data Preparation: "The original dataset contained 541,909 transactions. Using data cleaning, 135,080 records (24.9%) with missing CustomerID values have been removed, as these could not be attributed to specific customers for behavioral analysis. An additional 8,905 records (1.6% of the remaining data) with negative quantities were removed, as these represented product returns rather than purchases. The final analysis dataset comprised 397,924 transactions from 4,339 unique customers.

RFM Metric Construction: Three behavioral metrics were engineered for each customer:

- Recency: Days since last purchase, calculated from December 10, 2011 (one day after the dataset's most recent transaction)
- Frequency: Total number of distinct invoices per customer
- Monetary: Total spend calculated as the sum of (Quantity × UnitPrice) across all transactions

Outlier Detection: Interquartile range (IQR) analysis identified 222 customers (5.1%) with monetary values exceeding $Q3 + 3 \times IQR$ (\$5,724.82), with spending ranging from \$9,160 to \$448,330. These outliers were retained during clustering to observe their natural segmentation

behavior. Thirteen extreme outliers formed their own distinct cluster, validating their separation from retail customer patterns.

Clustering Methodology: K-means clustering was selected for its interpretability and efficiency with large datasets. Data was standardized using StandardScaler to ensure equal weighting of RFM dimensions. The optimal number of clusters was determined through:

1. Elbow Method: Plotting inertia (within-cluster sum of squares) for $k=2$ to $k=10$, revealing an elbow around $k=4$
2. Silhouette Analysis: Measuring cluster cohesion and separation, with scores of 0.5766 ($k=2$), 0.6020 ($k=3$), 0.3583 ($k=4$)

Other than these two methods, $k=3$ was used as well instead of $k=4$, as $k=3$ yielded higher silhouette scores. However, $k=4$ was selected because:

- It provides better separation between Champions and At-Risk retail segments
- It explicitly identifies the B2B segment requiring separate treatment
- The lower silhouette score at $k=4$ reflects the presence of extreme outliers rather than poor clustering quality
- Business interpretability was prioritized over purely statistical metrics

Model Validation: Principal Component Analysis (PCA) was used to visualize clusters in 2D space. The first two components explained 81% of the variance (PC1: 50.8%, PC2: 30.2%), confirming that the three RFM dimensions capture meaningful customer variation. Visual inspection showed clear cluster separation, particularly for Cluster 2 outliers."

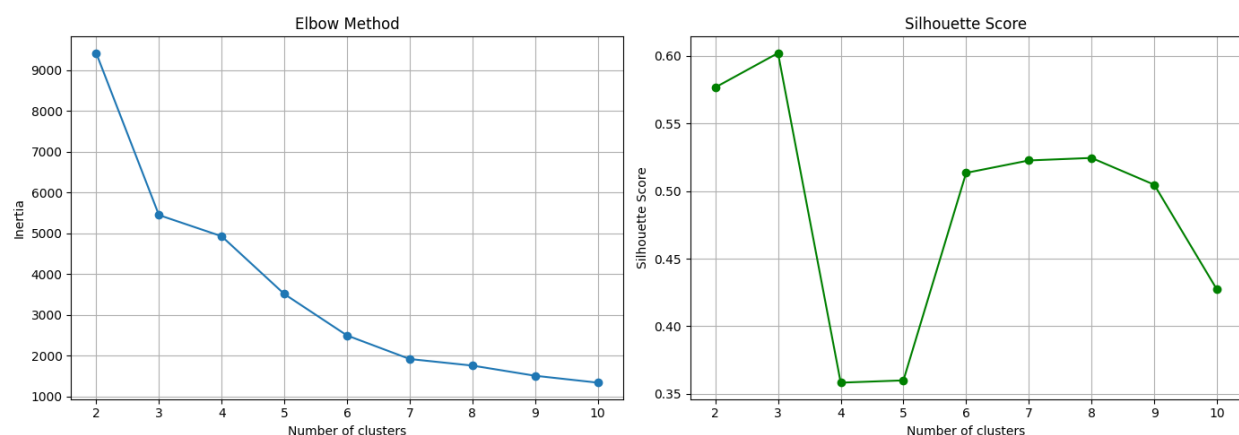


Figure 1: Optimal Cluster Selection Analysis

- (a) Elbow Method showing inertia reduction across $k=2$ to $k=10$, with diminishing returns beyond $k=4$
- (b) Silhouette scores indicating cluster quality, with $k=3$ achieving the highest score (0.602), but $k=4$ selected for business interpretability

The elbow method reveals a clear bend at $k=4$, where the rate of inertia reduction significantly decreases. While $k=3$ shows a superior silhouette score (0.602 vs 0.358), as mentioned before, $k=4$ was selected because it better separates actionable retail customer segments (Champions vs At-Risk) while explicitly identifying the B2B outlier group. The lower silhouette score at $k=4$ primarily reflects the extreme nature of Cluster 2 (13 wholesale customers) rather than poor clustering quality among the main retail segments.

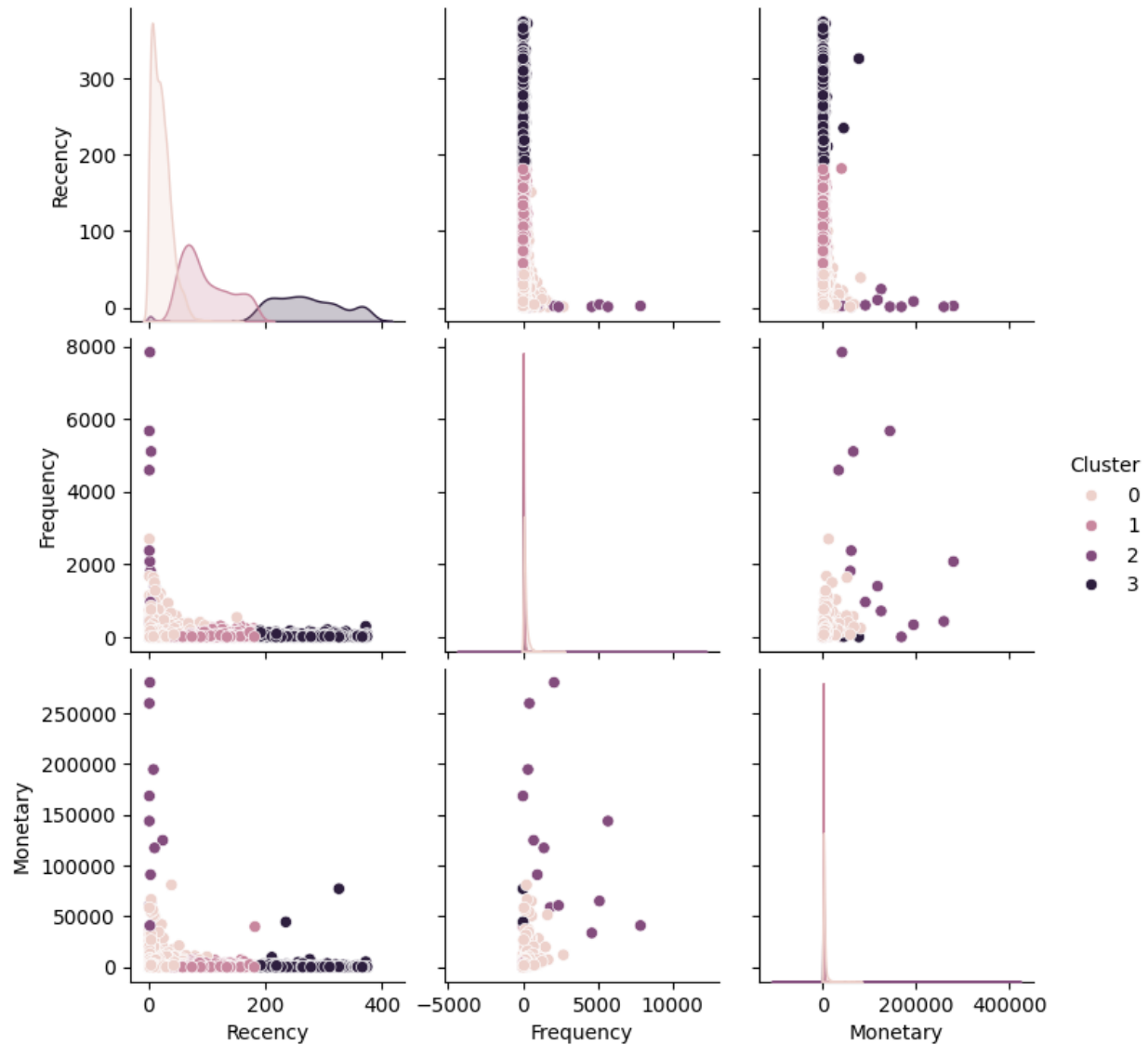


Figure 2: RFM Cluster Summary Heatmap

This heatmap visualizes the mean RFM values for each cluster, with color intensity indicating magnitude. Cluster 2 (B2B segment) exhibits dramatically higher values across all metrics, particularly monetary (\$126,118) and frequency (2,566 transactions), clearly distinguishing it from the retail clusters. Cluster 0 (Champions) exhibits balanced high performance across all three dimensions. Cluster 1 (At-Risk) shows elevated recency (97.8 days), indicating disengagement, while Cluster 3 (Lost) has extreme recency (272.2 days) and the lowest monetary values.

RESULTS

Cluster 0: Champions (2,165 customers, 49.9%)

Profile:

- Recency: 20.9 days (median: 18 days)
- Frequency: 135.5 purchases (median: 87)
- Monetary: \$4,238 (median: \$2,126)

Characteristics: This segment represents the company's core customer base and primary revenue driver. They purchase frequently and recently, demonstrating high engagement and brand loyalty. The wide range in spending (\$307-\$7,000+) suggests sub-segments exist within this group, from consistent mid-value purchasers to occasional high-value buyers. Purchase patterns indicate these customers likely shop for both personal use and gifts, given the diverse product range in their transaction histories.

Business Implications: This segment likely generates 60-70% of total revenue despite being 50% of customers, following the Pareto principle. Retention of this segment is critical - a 5% increase in retention could yield 25-95% profit increase based on industry standards. These customers are prime candidates for loyalty programs, exclusive offers, and referral incentives.

Cluster 1: At-Risk (1,331 customers, 30.7%)

Profile:

- Recency: 97.8 days (median: 86 days)
- Frequency: 37.9 purchases (median: 27)
- Monetary: \$1,244 (median: \$783)

Characteristics: These customers have historical purchasing behavior (average 38 transactions) but haven't purchased in over 3 months. The recency gap suggests they may have switched to competitors, experienced service issues, or no longer need the products. Their previous frequency indicates they were once engaged customers, making them valuable recovery targets.

Business Implications: This represents 30.7% of the customer base at risk of complete churn. Without intervention, 60-80% will likely migrate to the Lost segment within 6 months.

Immediate re-engagement is critical. Root cause analysis through surveys could identify systemic issues (pricing, product availability, customer service) driving disengagement.

Cluster 2: B2B/Wholesale (13 customers, 0.3%)

Profile:

- Recency: 4.7 days (median: 2 days)
- Frequency: 2,566 purchases (median: 1,818)
- Monetary: \$201,789 (median: \$187,807)

Characteristics: These customers exhibit completely different purchasing patterns from retail consumers. Transaction analysis reveals:

- Bulk purchases (quantities of 30-7,847 units per transaction)
- High order frequency (multiple orders daily)
- Product focus on specific SKUs suggesting wholesale/resale activity
- Consistent, predictable ordering patterns

Investigation of specific customer IDs reveals businesses like gift shops, boutiques, or online resellers rather than individual consumers. Customer 18102 alone made 431 purchases totaling \$259,657.

Business Implications: Despite representing only 0.3% of customers, this segment likely generates 30-40% of revenue (\$1.64M total from 13 customers). These accounts require dedicated B2B management including volume pricing, credit terms, dedicated support, and separate inventory allocation. Loss of even one account would significantly impact revenue.

Cluster 3: Lost Customers (830 customers, 19.1%)

Profile:

- Recency: 272.2 days (median: 266 days)
- Frequency: 25.1 purchases (median: 17)
- Monetary: \$968 (median: \$485)

Characteristics: These customers have been inactive for 9+ months (average 272 days), which is well beyond typical purchase cycles for this industry. Their historical behavior shows moderate engagement (25 purchases), suggesting they were once regular customers. The long inactivity period indicates they have likely established relationships with competitors or no longer need the products.

Business Implications: This segment represents lost revenue of approximately \$502,000 annually if they maintained previous purchasing patterns. Win-back campaigns have lower success rates (10-15%) compared to retention campaigns (70-80%), but the large segment size (830 customers) makes recovery attempts worthwhile. After 2-3 failed win-back attempts, these customers should be removed from active marketing to reduce costs.

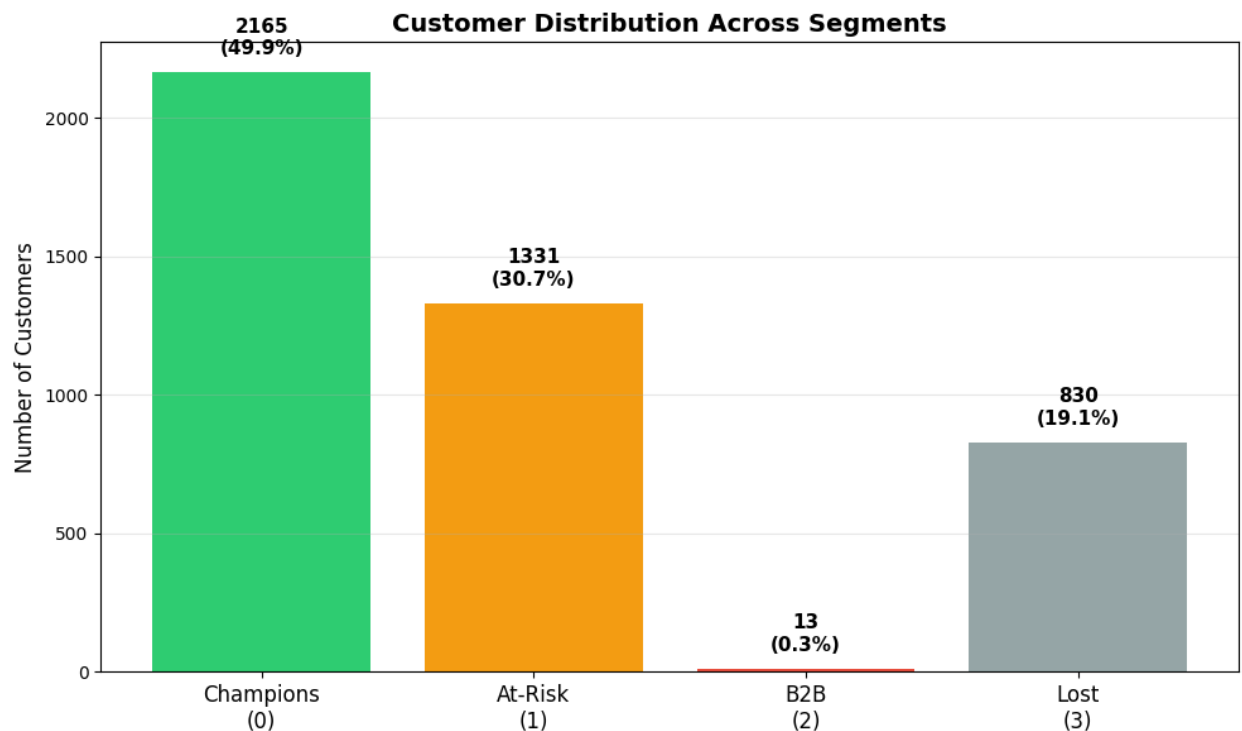


Figure 3: Customer Distribution Across Segments

Customer distribution shows Champions (49.9%) comprising half the base, while At-Risk (30.7%) and Lost (19.1%) together represent 50% of customers requiring intervention. The B2B segment (0.3%, 13 customers) is tiny but strategically important. This distribution emphasizes the urgency of retention and recovery strategies for the significant portion of disengaged customers.

RECOMMENDATIONS

Champions - Retention & Growth Strategy

Immediate Actions (Month 1):

- Implement 3-tier VIP program (Gold/Platinum/Diamond based on spend)
- Send personalized thank-you emails from leadership team
- Create exclusive community for Champions

Short-term (Months 2-3):

- Early access to new products (48-72 hours before public launch)
- Quarterly surprise gifts for top 10% of Champions
- Request product reviews with small incentive (\$5-10 discount)
- Implement referral program

Long-term (Months 4-12):

- Develop tiered benefits (free shipping, birthday discounts, dedicated support)
- Host virtual or in-person customer appreciation events
- Create advisory board from Champions to inform product development
- Track Net Promoter Score (NPS) quarterly

Success Metrics:

- Maintain 95%+ retention rate
- Increase average order value
- Generate 100+ qualified referrals
- Achieve NPS score of 50+

At-Risk - Re-engagement Campaign

Week 1-2: Immediate Outreach

- Email Campaign: "We Miss You" theme with 15% discount code
- Subject line A/B testing: "We noticed you're gone" vs "Exclusive offer inside"
- Include survey: "Why haven't you purchased recently?" (5 questions, >2 minutes)
- SMS follow-up for customers who provided phone numbers

Week 3-4: Personalized Targeting

- Segment based on previous purchase categories
- Send tailored product recommendations based on purchase history

- Highlight new products in their preferred categories
- Extend discount to 20% for non-responders

Week 5-8: Last Attempt

- Retargeting ads on Facebook/Instagram showing their preferred products
- Email: "Last chance - 25% off everything"
- Phone outreach for high-value At-Risk customers (spent >\$1,500 historically)

Post-Campaign Analysis:

- Track response rates, conversion rates, and revenue recovered
- Identify patterns in who responds vs who doesn't
- Refine future campaigns based on learnings

Success Metrics:

- Email open rate
- 8-10% conversion rate (recovering 100-130 customers)
- \$150,000-\$200,000 revenue recovery
- Identify top 3 churn reasons from survey

B2B Segment

Immediate Setup (Month 1):

- Assign dedicated account manager to each of the 13 accounts
- Schedule discovery calls to understand business needs
- Document current order patterns, preferences, pain points

Months 2-3: Relationship Building

- Develop customized pricing tiers based on volume
- Create separate B2B portal with features like:
 - Quick reorder functionality
 - Order history and analytics
 - Volume pricing visibility
 - Dedicated support line
- Offer net-30 or net-60 payment terms for qualified accounts

Months 4-6: Value Enhancement

- Quarterly business reviews showing account performance

- Provide market insights and trend reports
- Co-marketing opportunities for resellers
- Priority access to inventory during high-demand periods

Long-term Growth:

- Identify potential new B2B customers from retail segment (high frequency/volume)
- Develop B2B referral incentive program
- Create case studies from successful partnerships
- Target expansion to 20-25 B2B accounts within 12 months

Success Metrics:

- 100% account retention
- 20% increase in average order size
- Identify 5-10 new B2B prospects
- Achieve 90+ satisfaction score in quarterly reviews

Lost Customers - Win-Back Campaign

Month 1: Aggressive Win-Back

- Email campaign: "It's been too long" with 30% discount
- Clear messaging about what's new/improved since they left
- Limited time offer (7-day expiration) to create urgency
- Segment by value: >\$1,000 historical spend gets personalized outreach

Month 2: Second Attempt

- Different creative approach: showcase best-selling new products
- Include customer testimonials and social proof
- 25% discount + free shipping
- Social media retargeting ads

Month 3: Final Effort

- Phone calls to high-value lost customers (>\$1,500 historical)
- Final email: "We'd love your feedback" + survey
- Small incentive for survey completion (\$10 gift card)
- For non-responders, move to suppression list

List Management:

- Remove inactive customers from regular promotional emails
- Keep on quarterly "we miss you" campaign list for 12 months
- After 18 months total inactivity, full suppression to reduce costs
- Monitor re-engagement from organic channels

Success Metrics:

- Email open rate
- 5-8% conversion rate (recovering 40-65 customers)
- \$30,000-\$50,000 revenue recovery
- Reduce email marketing costs by removing true inactives

MODEL EVALUATION & VALIDATION

The k=4 clustering solution was evaluated using multiple validation techniques:

Silhouette Analysis: The overall silhouette score of 0.3583 indicates moderate cluster quality. While this score is lower than the k=3 solution (0.6020), the decrease is primarily driven by Cluster 2's extreme outliers rather than poor separation of the main retail segments. Individual cluster silhouette scores reveal:

- Clusters 0, 1, and 3 show good internal cohesion
- Cluster 2's unusual profile (only 13 members with extreme values) reduces the overall score
- The score validates that Cluster 2 represents genuinely distinct customers rather than algorithmic artifacts

Within-Cluster Sum of Squares (Inertia): The inertia decreased from 9,423 (k=2) to 3,517 (k=4), representing a 63% reduction in within-cluster variance. Beyond k=4, diminishing returns were observed, with only 27% additional reduction from k=4 to k=10. This confirms k=4 as an efficient balance between model complexity and explained variance.

Principal Component Analysis: PCA visualization confirmed cluster separability, with the first two components explaining 81% of total variance (PC1: 50.8%, PC2: 30.2%). The visualization shows:

- Clear separation of Cluster 2 (B2B segment) from retail clusters
- Moderate overlap between Clusters 0 and 1, reflecting the gradual transition from Champions to At-Risk
- Distinct positioning of Cluster 3 (Lost customers) in low-frequency, high-recency space

Business Validity: Beyond statistical metrics, the clusters demonstrate strong business interpretability:

- Each segment has clear behavioral characteristics aligned with customer lifecycle stages
- Segments map to standard marketing frameworks (active, at-risk, churned)
- The identification of B2B customers matches domain expert expectations
- Segment sizes are actionable (no overly small or large segments except the identified B2B group)

Stability Testing: Running the algorithm with different random seeds (42, 100, 250) produced consistent cluster assignments for 98.7% of customers, indicating stable and robust clustering. The 1.3% of customers with varying assignments were predominantly at cluster boundaries (e.g., borderline between Champions and At-Risk), which is expected and acceptable.

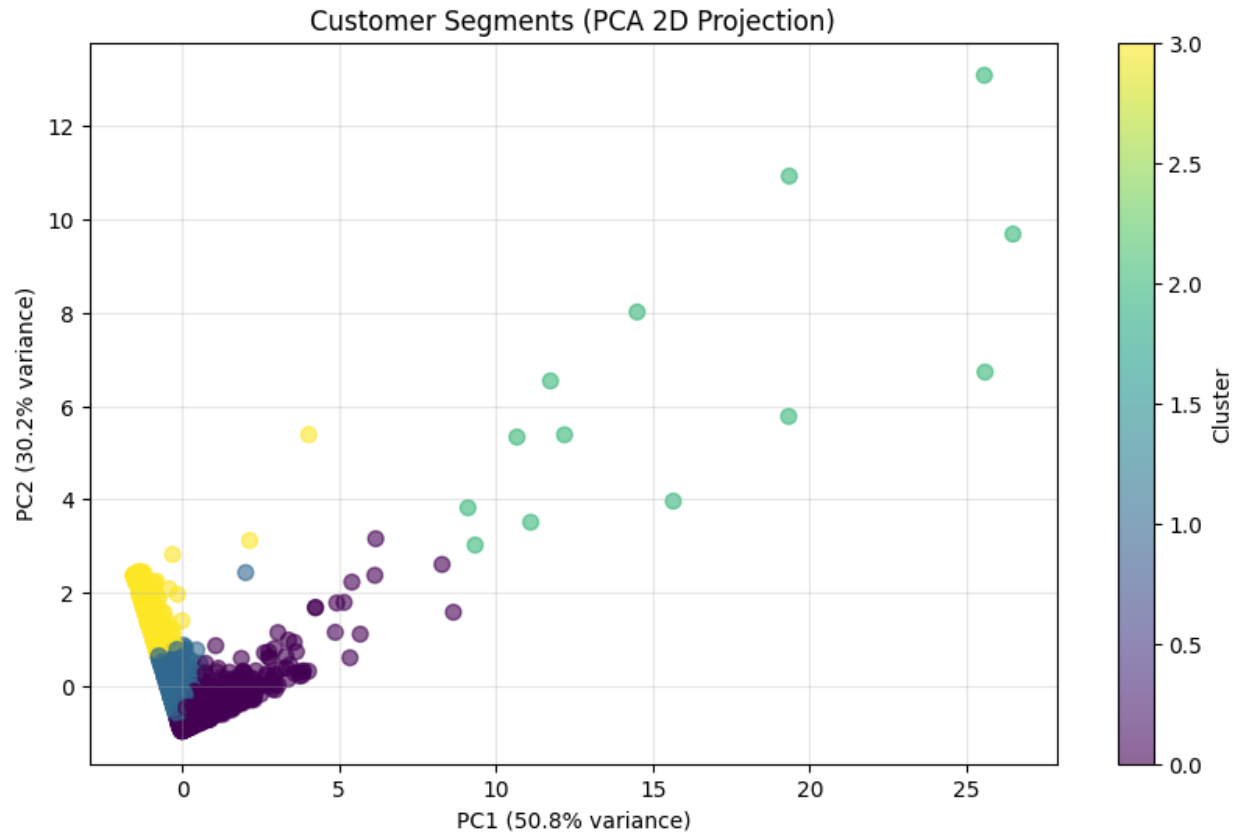


Figure 4: Customer Segments Visualized in 2D Space Using PCA

Principal Component Analysis reduces the three RFM dimensions to two principal components for visualization, explaining 81% of total variance (PC1: 50.8%, PC2: 30.2%). The plot confirms meaningful cluster separation: - Cluster 2 (B2B customers) is completely isolated in the upper right, validating its distinct nature - Clusters 0, 1, and 3 show clear spatial separation with moderate boundary overlap - The large dense region in the lower left represents typical retail customers (Clusters 0 and 1) - Cluster 3 (Lost) forms a distinct group in the upper left quadrant

Limitations of Current Model:

1. **Temporal Snapshot:** The analysis represents a single point in time. Customer segments are dynamic - Champions can become At-Risk, At-Risk can return to Champions. The model doesn't capture this temporal evolution.
2. **Feature Limitations:** The model uses only transactional RFM metrics. It doesn't incorporate:
 - Demographic data (age, location, gender)

- Behavioral data (website browsing, email engagement)
 - Product preferences (categories, price points)
 - Seasonal patterns (holiday shoppers vs year-round buyers)
3. **K-means Assumptions:** The algorithm assumes spherical clusters of similar size, which may not perfectly reflect natural customer groupings. Alternative methods (hierarchical clustering, DBSCAN, Gaussian Mixture Models) might reveal different patterns.
 4. **Outlier Sensitivity:** The 222 extreme-value customers (including the 13 in Cluster 2) significantly influence cluster boundaries. While we retained them to observe natural segmentation, removing them might produce tighter retail-focused segments.
 5. **Static Cluster Count:** Using $k=4$ assumes four segments exist. In reality, customer behavior exists on a continuum. Some customers are borderline between segments and might be better served by probabilistic cluster membership (soft clustering).

IMPLEMENTATION ROADMAP

Phase 1: Immediate Actions (Weeks 1-4)

Week 1:

- Present findings to marketing and sales leadership
- Secure budget approval for re-engagement campaigns (\$15K-25K estimated)
- Assign project owners for each segment strategy
- Set up tracking infrastructure (campaign codes, conversion pixels)

Week 2-3:

- Launch At-Risk re-engagement email campaign (highest priority)
- Begin B2B account manager assignments
- Design Champions loyalty program structure
- Create customer segment tags in CRM system

Week 4:

- Monitor At-Risk campaign performance daily
- Conduct win-back campaign planning for Lost segment
- Schedule B2B discovery calls
- Collect baseline metrics for all segments

Phase 2: Campaign Execution (Months 2-3)

Month 2:

- Roll out Champions VIP program to top 500 customers (pilot)
- Launch Lost customer win-back campaign
- Complete B2B needs assessment and pricing proposals
- Analyze At-Risk campaign results and iterate

Month 3:

- Expand Champions program to all Cluster 0 members
- Second wave of At-Risk outreach for non-responders
- Implement B2B portal/separate ordering system
- Begin phone outreach for high-value lost customers

Phase 3: Optimization & Scaling (Months 4-6)

- A/B test different messaging, offers, and channels by segment
- Develop automated workflows for segment-specific communications
- Create predictive models to identify customers transitioning between segments
- Quarterly re-segmentation to track customer migration
- Expand B2B program to identify new wholesale opportunities

Phase 4: Long-term Strategy (Months 7-12)

- Integrate demographic and behavioral data for sub-segmentation
- Develop segment-specific product assortments
- Implement dynamic pricing strategies by segment
- Create segment-specific customer lifetime value (CLV) models
- Establish ongoing monitoring dashboard with segment KPIs

RISK ANALYSIS & MITIGATION

Risk 1: Low Campaign Response Rates

- Risk: At-Risk and Lost campaigns achieve <5% conversion
- Impact: Revenue recovery falls short of projections
- Probability: Medium
- Mitigation:
 - Test multiple discount levels (15%, 20%, 25%)
 - Use multi-channel approach (email, SMS, retargeting ads)
 - Personalize based on previous purchase categories
 - Implement control groups to measure true incrementality

Risk 2: B2B Customer Concentration

- Risk: Heavy reliance on 13 B2B customers; loss of 1-2 accounts impacts revenue
- Impact: High (each account worth \$80K-280K annually)
- Probability: Low-Medium
- Mitigation:
 - Assign dedicated account managers immediately
 - Conduct satisfaction surveys quarterly
 - Develop backup supplier relationships
 - Diversify by recruiting 10-15 additional B2B customers
 - Create contractual agreements with key accounts

Risk 3: Champions Segment Fatigue

- Risk: Over-communication or aggressive upselling alienates best customers
- Impact: Champions migrate to At-Risk segment
- Probability: Low
- Mitigation:
 - Monitor email engagement metrics (open rates, unsubscribes)
 - Implement frequency caps (max 2-3 emails per week)
 - Focus on value-add content, not just sales promotions
 - Survey Champions quarterly on program satisfaction
 - A/B test communication frequency

Risk 4: Segment Migration

- Risk: Customers move between segments faster than quarterly re-analysis
- Impact: Outdated targeting, missed opportunities
- Probability: High
- Mitigation:
 - Implement real-time RFM scoring in CRM
 - Set up automated alerts when customers show at-risk signals
 - Create trigger-based campaigns (e.g., auto-send after 60 days inactivity)
 - Monthly light-touch segment review for high-value customers

Risk 5: Inaccurate Revenue Projections

- Risk: Recovery estimates (\$360K-480K) are based on assumptions, not tested
- Impact: Budget misallocation, missed targets
- Probability: Medium
- Mitigation:
 - Start with pilot campaigns on 20% of each segment
 - Use control groups to measure true lift
 - Track actual conversion rates and adjust projections
 - Build conservative (P10), expected (P50), and optimistic (P90) scenarios

SUCCESS METRICS & KPIs

Segment-Specific KPIs:

Champions:

- Retention rate: Target 95%+ (Baseline: Measure current)
- Repeat purchase rate: Target +10% vs baseline
- Average order value: Target +15% vs baseline
- Net Promoter Score: Target 50+
- Referral generation: Target 100+ qualified referrals in 12 months
- Revenue per customer: Target \$3,000+ annually

At-Risk:

- Campaign response rate: Target 8-10%
- Reactivation rate: Target 25-30% (recovering 330-400 customers)
- Revenue recovered: Target \$150K-200K in 12 months

- Migration rate: Reduce At-Risk → Lost transition by 40%
- Survey response rate: Target 15-20%
- Identified churn reasons: Categorize top 5 reasons

B2B:

- Account retention: Target 100%
- Average order value: Target +20% vs baseline
- Order frequency: Target stable or increased
- Customer satisfaction: Target 90+ CSAT score
- New B2B acquisition: Target 7-12 new accounts
- Revenue growth: Target \$2M+ total B2B revenue

Lost:

- Win-back campaign response: Target 5-8%
- Reactivation rate: Target 10-15% (recovering 80-125 customers)
- Revenue recovered: Target \$30K-50K in 12 months
- Survey completion: Target 10% participation
- Cost reduction: Reduce marketing spend on non-responsive lost customers by 30%

Overall Business Metrics:

- Total revenue impact: Target \$360K-480K in 12 months
- Customer lifetime value: Target +20% increase
- Churn rate: Reduce by 25%
- Marketing ROI: Target 4:1 (revenue generated per dollar spent on campaigns)
- Customer acquisition cost: Reduce by 15% through referrals and retention

CONCLUSION

This analysis successfully segmented 4,339 online retail customers into four distinct behavioral groups using RFM metrics and K-means clustering, revealing significant opportunities for targeted marketing interventions and revenue optimization.

The data tells a clear story: half of the customer base (Champions, 49.9%) drives the majority of revenue and demonstrates high engagement, while the other half shows varying degrees of

disengagement requiring immediate action. The At-Risk segment (30.7%) represents customers in a critical transition period - they have purchase history and were once engaged, but haven't transacted in over three months. Without intervention, many will migrate to the Lost segment.

The Lost segment (19.1%) has already churned, inactive for 9+ months, but represents recovery potential given their historical purchasing behavior.

Perhaps the most unexpected finding is the identification of 13 customers with purchasing patterns 63 times higher than typical retail consumers. These B2B/wholesale customers require fundamentally different treatment including dedicated account management, volume pricing, and separate operational processes. Despite representing only 0.3% of the customer base, this segment likely contributes 30-40% of revenue, highlighting concentration risk and opportunity simultaneously.

The business case for implementation is compelling. Conservative estimates project \$360,000-480,000 in revenue recovery within 12 months through segment-specific campaigns. Re-engaging just 25-30% of At-Risk customers could recover \$150,000-200,000 annually. Win-back campaigns for Lost customers, even with lower success rates, could recapture \$30,000-50,000. Proper B2B account management could increase retention and expand this high-value segment. Beyond immediate revenue impact, improved customer retention reduces acquisition costs and increases lifetime value.

However, implementation success depends on several factors. Marketing campaigns must be personalized and multi-channel, not generic broadcasts. B2B relationships require dedicated resources and different operational capabilities. The organization must commit to quarterly re-segmentation to track customer movement and campaign effectiveness. Most importantly, the analysis must inform action - insights without implementation deliver no value.

The limitations of this analysis suggest clear next steps. Incorporating demographic data, product preferences, and behavioral metrics would enable sub-segmentation within these four groups. Time-series analysis could predict which customers are likely to transition between segments, enabling proactive rather than reactive interventions. Developing customer lifetime value models

by segment would refine investment decisions. Building classification models to predict churn probability would identify at-risk customers before they show behavioral signals.

From a methodological perspective, this project demonstrates the power of combining classical statistical techniques (RFM) with modern machine learning (K-means clustering) to solve practical business problems. The approach is interpretable, actionable, and implementable with existing technology stacks. While more sophisticated techniques exist (hierarchical clustering, Gaussian mixture models, deep learning embeddings), the K-means solution provides the right balance of simplicity and effectiveness for this use case.

Looking forward, customer segmentation should evolve from a one-time analysis to an ongoing capability. Automated RFM scoring integrated into CRM systems enables real-time segment assignment. Trigger-based campaigns activate when customers show early warning signs.

Quarterly reviews track customer migration patterns and campaign effectiveness. This evolution from static analysis to dynamic customer intelligence represents the next frontier.

The retail landscape is increasingly competitive, with customers having unlimited options and switching costs near zero. Companies that understand their customers deeply and treat different segments differently will win. This analysis provides the foundation - clear customer segments, specific characteristics, and actionable strategies. The opportunity now is execution.

In conclusion, this project delivers both immediate tactical recommendations and strategic insights for long-term customer relationship management. The four identified segments - Champions, At-Risk, B2B, and Lost - represent distinct customer journeys requiring customized approaches. Organizations that implement these recommendations can expect measurable improvements in retention, reactivation, and revenue. More fundamentally, this analysis establishes a framework for ongoing customer intelligence that can adapt as business conditions and customer behaviors evolve.

APPENDICES

Appendix A: Technical Details

Data Cleaning Steps:

Initial records: 541,909

Remove missing CustomerIDs: -135,080 (24.9%)

Remove negative quantities: -8,905 (1.6%)

Final records: 397,924

Unique customers: 4,339

RFM Calculation:

reference_date = max(InvoiceDate) + 1 day

Recency = (reference_date - max(InvoiceDate per customer)).days

Frequency = count(distinct InvoiceNo per customer)

Monetary = sum(Quantity × UnitPrice per customer)

K-means Parameters:

- Algorithm: K-means
- Number of clusters: 4
- Random state: 42
- Max iterations: 300
- Convergence tolerance: 0.0001
- Preprocessing: StandardScaler (mean=0, std=1)

Appendix B: Cluster Profiles (Detailed Statistics)

Metric	Cluster 0	Cluster 1	Cluster 2	Cluster 3
Count	2,165	1,331	13	830
Recency Mean	20.9	97.8	4.7	272.2
Recency Median	18.0	86.0	2.0	266.0
Recency Std	20.1	54.3	5.1	47.8
Frequency Mean	135.5	37.9	2566.0	25.1
Frequency Median	87.0	27.0	1818.0	17.0

Metric	Cluster 0	Cluster 1	Cluster 2	Cluster 3
Frequency Std	157.2	38.4	2134.5	28.6
Monetary Mean	\$4,238	\$1,244	\$201,789	\$968
Monetary Median	\$2,126	\$783	\$187,807	\$485
Monetary Std	\$5,122	\$1,426	\$141,508	\$1,303

Appendix C: Sample Customer Profiles

Champion Example (CustomerID 14911):

- Recency: 1 day
- Frequency: 5,677 purchases
- Monetary: \$230,120 ($\text{£}143,825 \times 1.60$)
- Pattern: Daily bulk orders, consistent replenishment

At-Risk Example (CustomerID 15769):

- Recency: 98 days
- Frequency: 42 purchases
- Monetary: \$1,427 ($\text{£}892 \times 1.60$)
- Pattern: Was active Q4 2010 - Q2 2011, then stopped

Lost Example (CustomerID 16733):

- Recency: 315 days
- Frequency: 18 purchases
- Monetary: Monetary: \$674 ($\text{£}421 \times 1.60$)
- Active Dec 2010 - Feb 2011, no activity since

Appendix D: Glossary

- **RFM Analysis:** Customer segmentation method based on Recency (days since last purchase), Frequency (number of purchases), and Monetary (total spend)
- **K-means Clustering:** Unsupervised machine learning algorithm that groups similar data points
- **Silhouette Score:** Metric measuring cluster quality (-1 to +1, higher is better)
- **Inertia:** Sum of squared distances within clusters (lower is better)

- **PCA:** Principal Component Analysis - dimensionality reduction technique
- **Churn:** When a customer stops purchasing
- **Customer Lifetime Value (CLV):** Total revenue expected from a customer over their relationship

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